Rental market ETL: one page

Flow: Raw CSV, PySpark transforms, partitioned Parquet (state, year), DuckDB, dbt marts, orchestrated by Airflow. Optional second runtime on Databricks Free Edition (ADR 0001).

Data quality: local check to Databricks expectation

Local check	Databricks expectation	Action
null % of median_rent <= 10%	median_rent IS NOT NULL (silver); null_pct <= 0.1 (aggregate)	drop / fail
row count >= 100	row_count >= 100	fail
rent in [0, 150000]	median_rent BETWEEN 0 AND 150000	fail
unique (RegionID, month)	duplicate_count = 0	warn

Parity (results/databricks_parity.json, run 2026-11-07)

Local PySpark 3.5.4 vs Databricks serverless 4.2.0: **diffs is empty**. Rows: bronze 8,340; silver_long, silver_zori and gold 469,420 each. The whole-row checksum matches on every layer.

Failure drill (tests/fixtures/bad_zori.csv)

A CSV with a null rent, a duplicate key and one rent of 200,000. Null and duplicate are removed before DQ; the range check fails, locally and on Databricks (gold_zori rent_in_range failed 1; dq_gold_metrics skipped due to upstream failure).

Timings (wall-clock, Free Edition serverless)

Real CSV: 136 s and 155 s. Failing fixture: 945 s (the update retried 6 times).

Not shown in this repo

Terraform/CloudFormation, Datadog/CloudWatch/Grafana, Scala, Glue, EMR, Redshift, Step Functions. S3 code paths exist but were not run end to end.